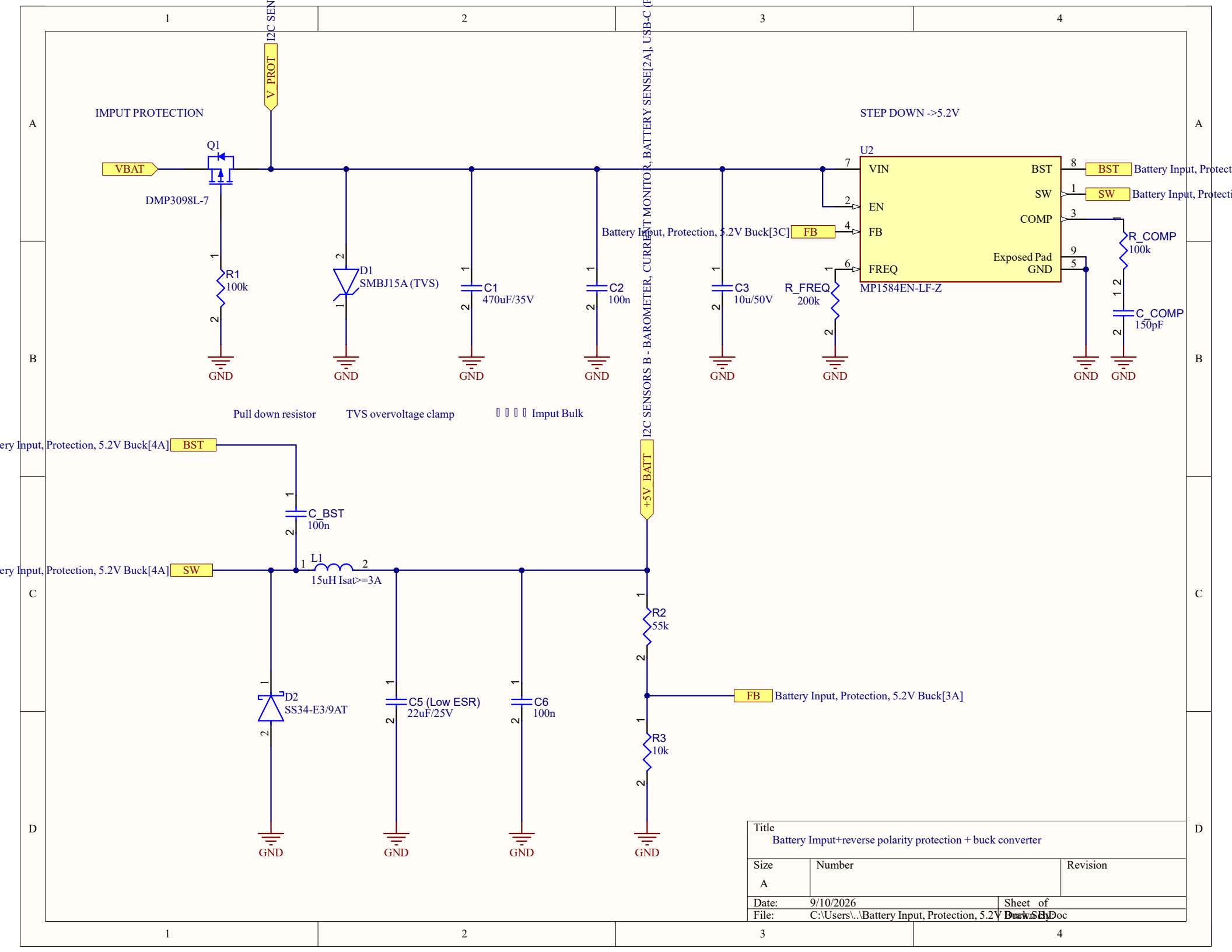
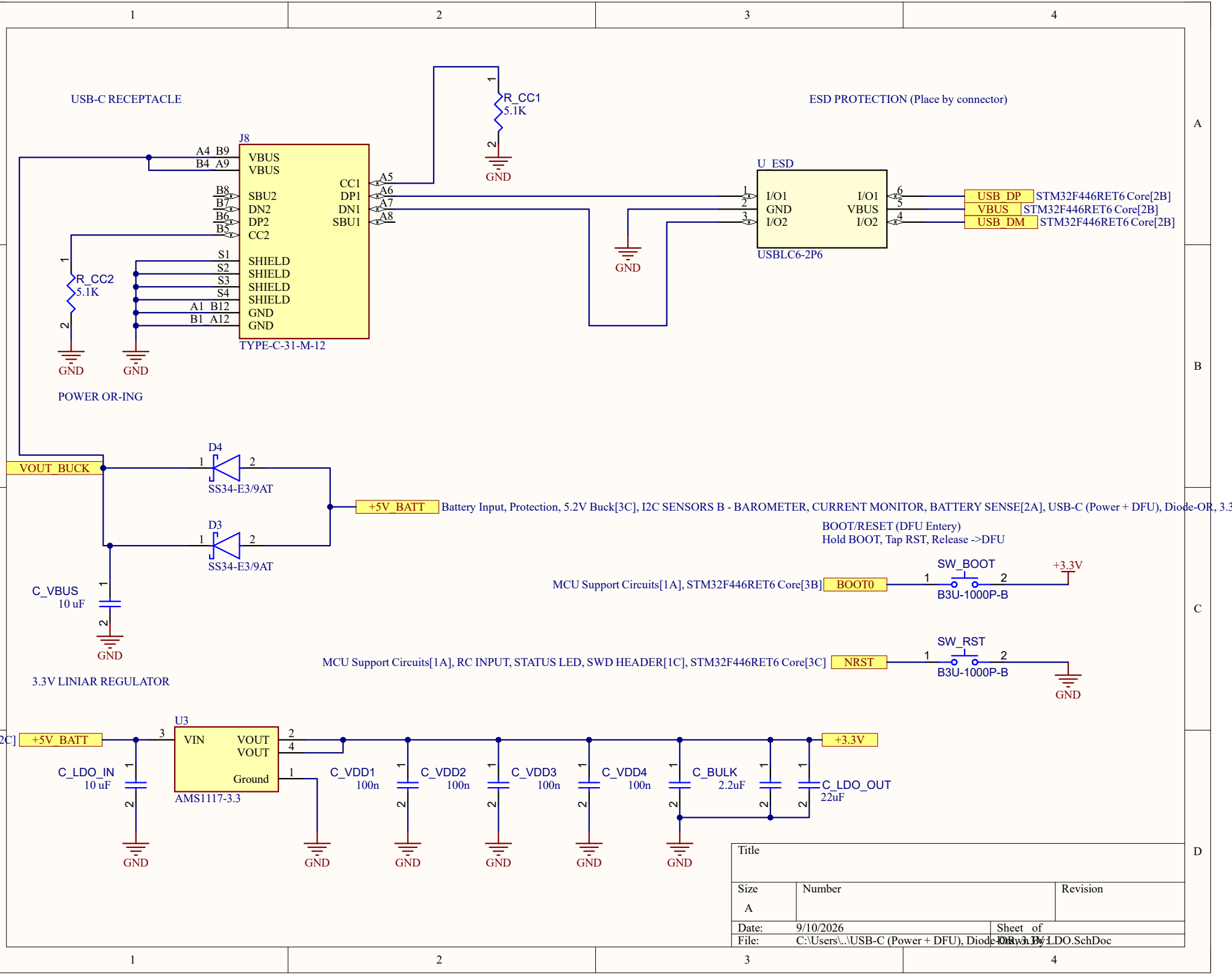
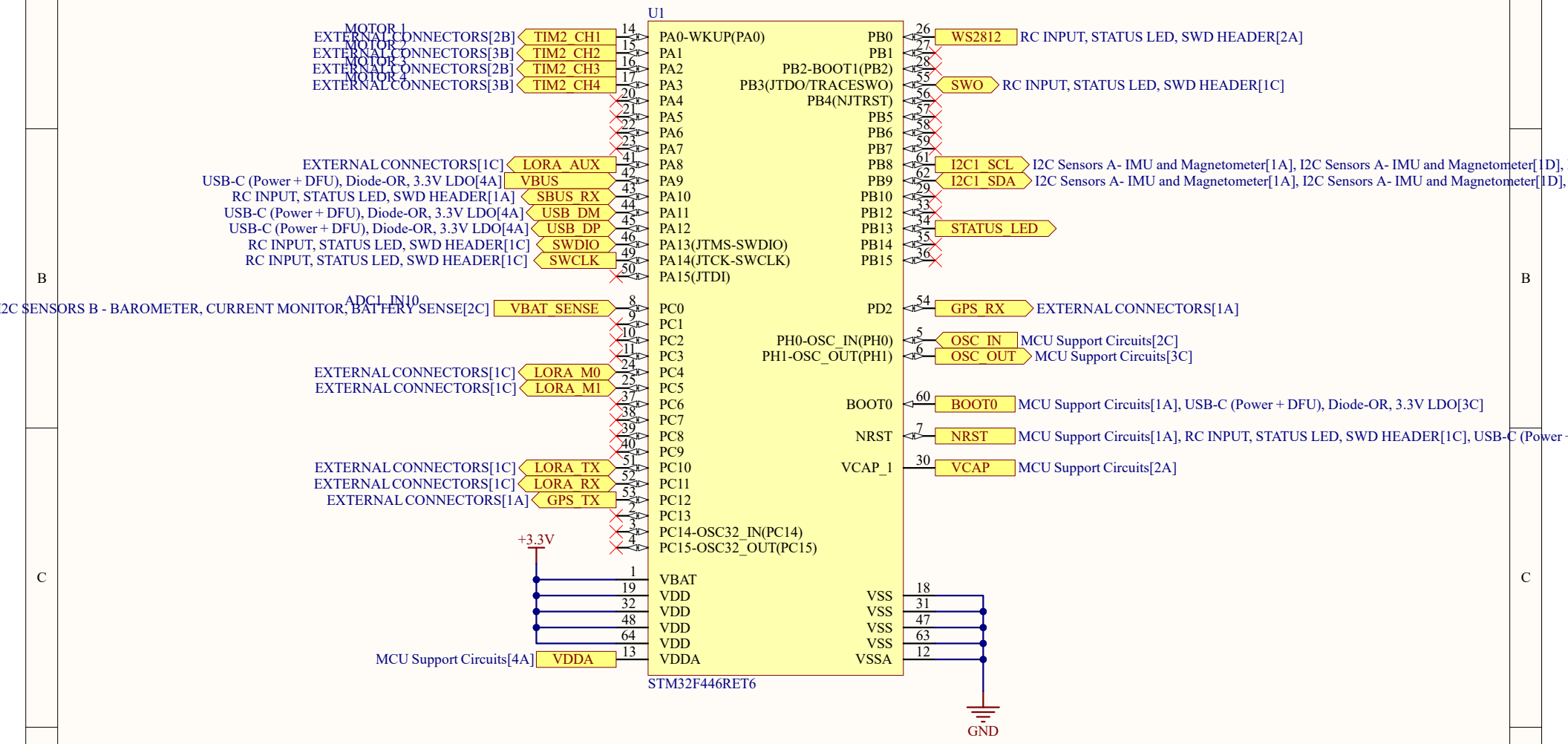
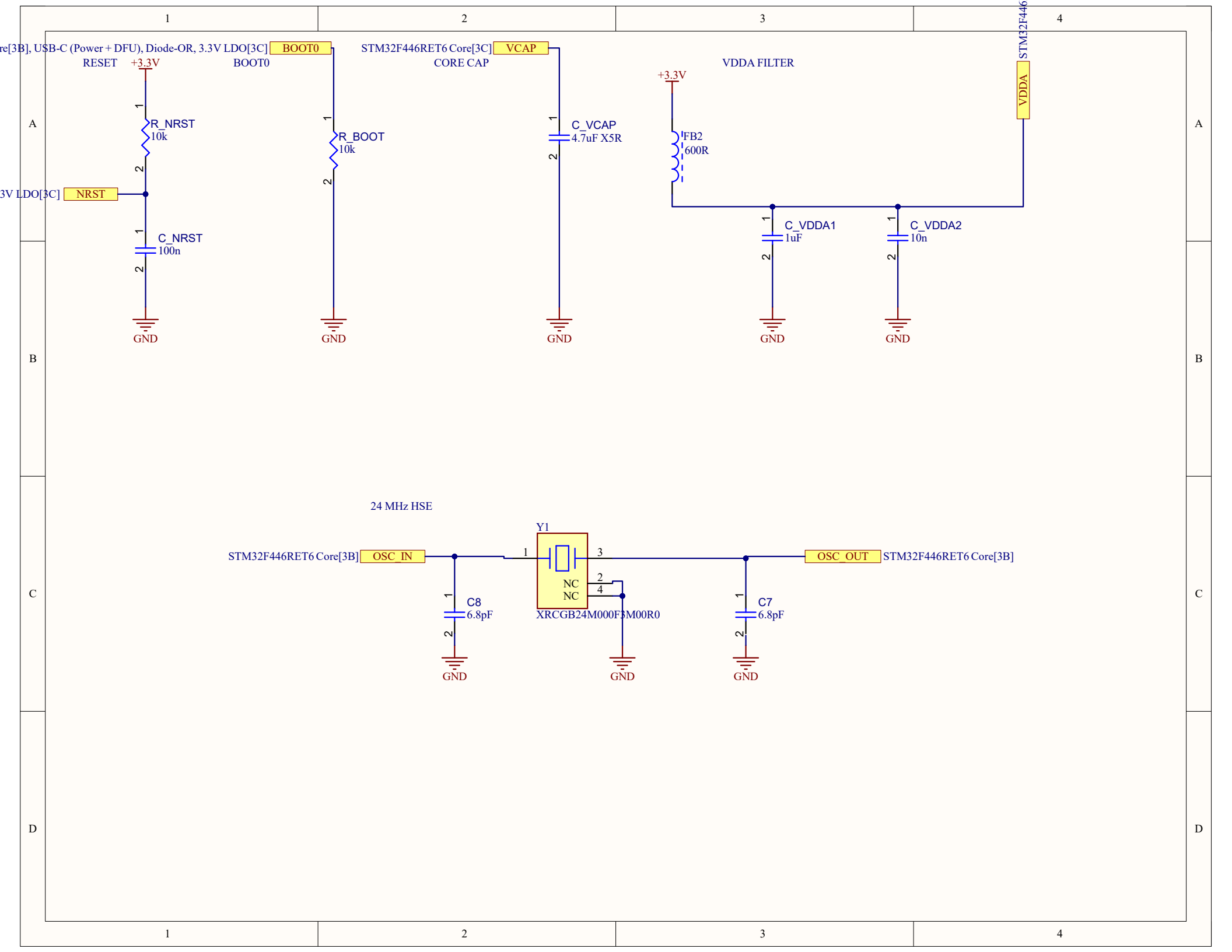

I2C SENSORS A - PROXIMITY, LIGHT SENSOR, V_PROT
I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE[2A], USB-C [F
IMPUT PROTECTION
STEP DOWN ->5.2V
VBAT
Q1
DMP3098L-7
R1
100k
D1
SMBJ15A (TVS)
C1
470uF/35V
C2
100n
C3
10u/50V
U2
VIN
EN
FB
FREQ
BST
SW
COMP
Exposed Pad
GND
MP1584EN-LF-Z
R_FREQ
200k
R_COMP
100k
C_COMP
150pF
Battery Input, Protection, 5.2V Buck[3C]
FB
BST
SW
Battery Input, Protect
Battery Input, Protecti
Pull down resistor
TVS overvoltage clamp
Imput Bulk
Battery Input, Protection, 5.2V Buck[4A]
C_BST
100n
L1
15uH Isat>=3A
+5V_BATT
R2
55k
R3
10k
D2
SS34-E3/9AT
C5 (Low ESR)
22uF/25V
C6
100n
Battery Input, Protection, 5.2V Buck[3A]
Title
Battery Imput+reverse polarity protection + buck converter
Size
A
Number
Revision
Date:
9/10/2026
Sheet of
File:
C:\Users\..\Battery Input, Protection, 5.2V Buck.SchDoc
Drawn By:

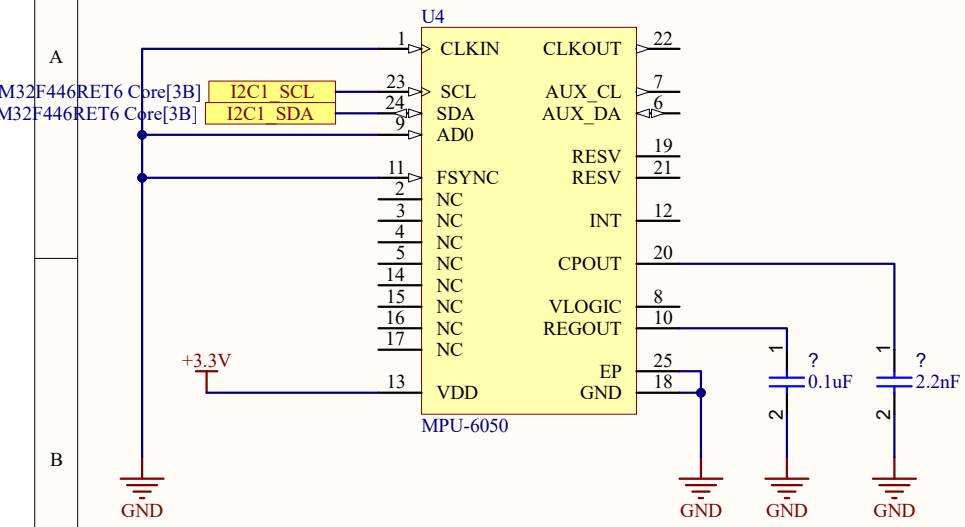

USB-C RECEPTACLE
J8
A4 B9
B4 A9
VBUS
VBUS
B8
B7
B6
B5
SBU2
DN2
DP2
CC2
CC1
DP1
DN1
SBU1
A5
A6
A7
A8
S1
S2
S3
S4
A1 B12
B1 A12
SHIELD
SHIELD
SHIELD
SHIELD
GND
GND
TYPE-C-31-M-12
R_CC1
5.1K
R_CC2
5.1K
GND
GND
GND
GND
ESD PROTECTION (Place by connector)
U_ESD
I/O1
GND
I/O2
I/O1
VBUS
I/O2
USBLC6-2P6
USB_DP
VBUS
USB_DM
STM32F446RET6 Core[2B]
STM32F446RET6 Core[2B]
STM32F446RET6 Core[2B]
POWER OR-ING
VOUT_BUCK
D4
SS34-E3/9AT
D3
SS34-E3/9AT
+5V_BATT
Battery Input, Protection, 5.2V Buck[3C], I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE[2A], USB-C (Power + DFU), Diode-OR, 3.3
BOOT/RESET (DFU Entery)
Hold BOOT, Tap RST, Release ->DFU
MCU Support Circuits[1A], STM32F446RET6 Core[3B]
BOOT0
SW_BOOT
B3U-1000P-B
+3.3V
MCU Support Circuits[1A], RC INPUT, STATUS LED, SWD HEADER[1C], STM32F446RET6 Core[3C]
NRST
SW_RST
B3U-1000P-B
GND
C_VBUS
10 uF
GND
3.3V LINIAR REGULATOR
V LDO[2C]
+5V_BATT
U3
VIN
VOUT
VOUT
Ground
AMS1117-3.3
C_LDO_IN
10 uF
C_VDD1
100n
C_VDD2
100n
C_VDD3
100n
C_VDD4
100n
C_BULK
2.2uF
C_LDO_OUT
22uF
+3.3V
GND
GND
GND
GND
GND
GND
GND
Title
Size
A
Number
Revision
Date:
9/10/2026
Sheet of
File:
C:\Users\..\USB-C (Power + DFU), Diode-OR, 3.3V LDO.SchDoc
Drawn By:



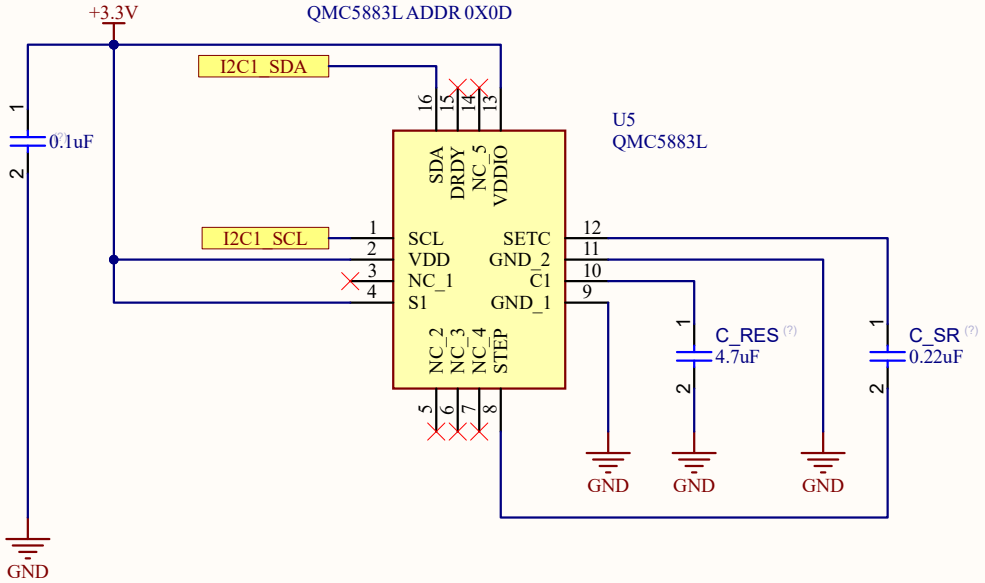

| | | |
|---|---|---|
| Title | | |
| Size: A | Number | Revision |
| Date: 9/10/2026 | | Sheet of |
| File: C:\Users\..\STM32F446RET6 Core.SchDoc | | Drawn By: |


re[3B], USB-C (Power + DFU), Diode-OR, 3.3V LDO[3C]
BOOT0
STM32F446RET6 Core[3C]
VCAP
VDDA FILTER
STM32F446
RESET
+3.3V
BOOT0
CORE CAP
+3.3V
VDDA
R_NRST
10k
R_BOOT
10k
C_VCAP
4.7uF X5R
FB2
600R
3V LDO[3C]
NRST
C_NRST
100n
C_VDDA1
1uF
C_VDDA2
10n
GND
GND
GND
GND
GND
24 MHz HSE
Y1
STM32F446RET6 Core[3B]
OSC_IN
NC
NC
OSC_OUT
STM32F446RET6 Core[3B]
C8
6.8pF
XRCGB24M000F3M00R0
C7
6.8pF
GND
GND
GND

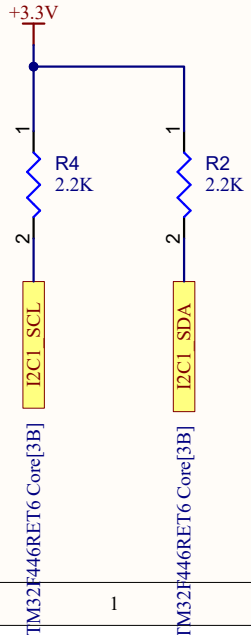
MPU-6050 ADDR 0X68



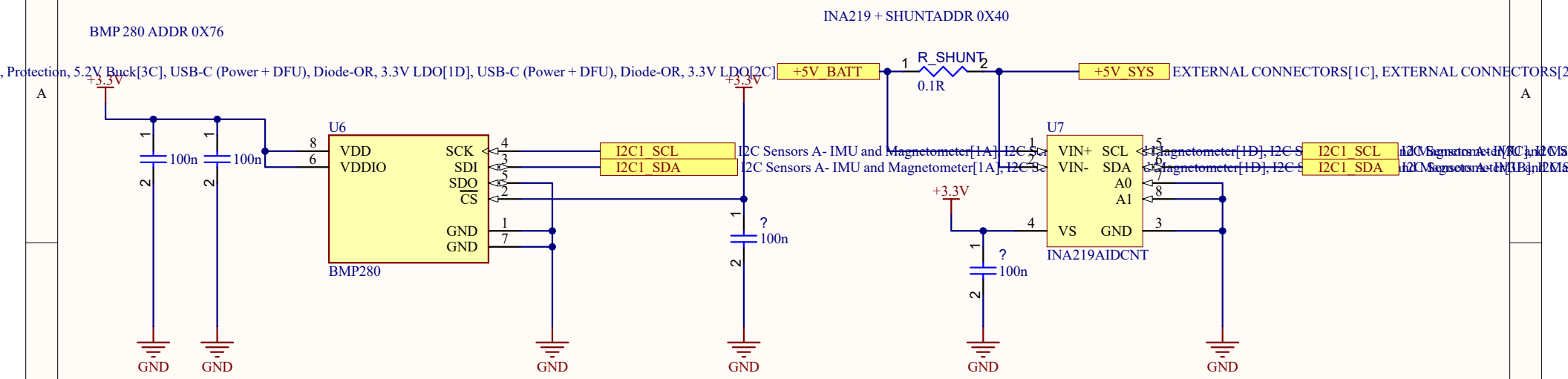

QMC5883L ADDR 0X0D

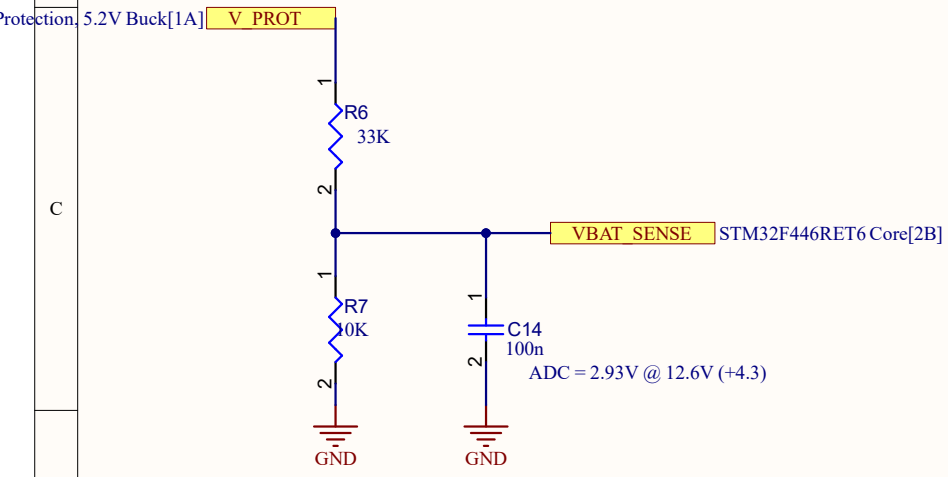



BUS PULL-UPS

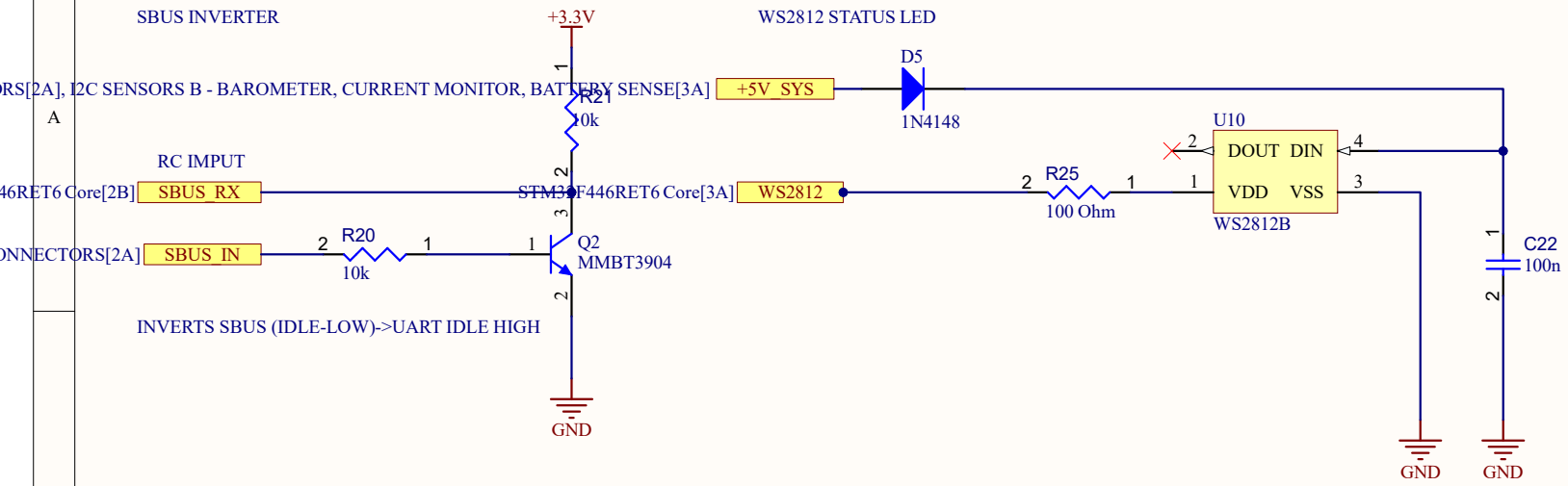


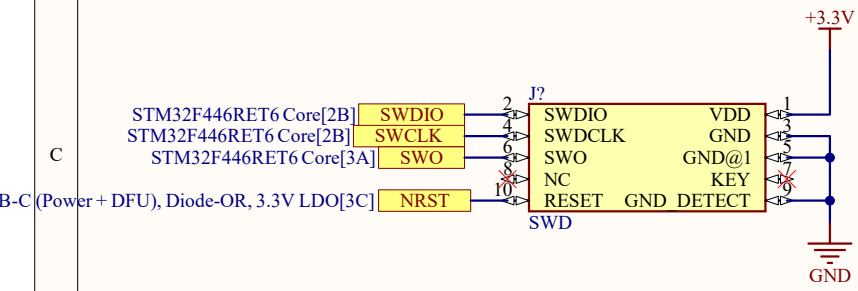

| | | |
|---|---|---|
| Title | | |
| Size: A | Number | Revision |
| Date: 9/10/2026 | | Sheet of |
| File: C:\Users\..\I2C Sensors A- IMU and Magnetometer.SchDoc | | Drawn By: |



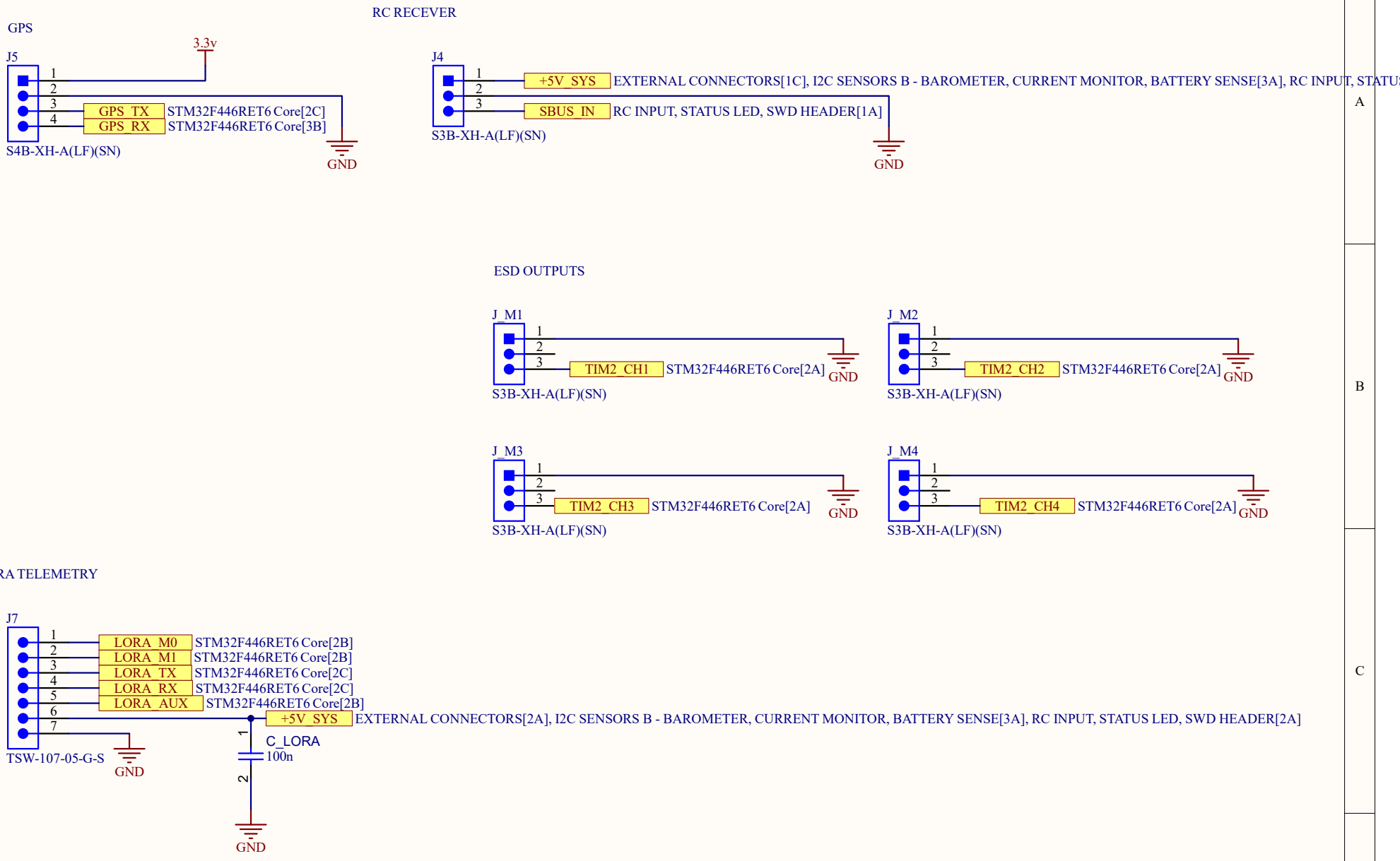




| | | |
|---|---|---|
| Title | | |
| Size: A | Number | Revision |
| Date: 9/10/2026 | | Sheet of |
| File: C:\Users\..\I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE.... | | Drawn By: |




| | | |
|---|---|---|
| Title | | |
| Size A | Number | Revision |
| Date: 9/10/2026 | | Sheet of |
| File: C:\Users\..\RC INPUT, STATUS LED, SWD HEADER.SchDoc | | Drawn By: |


RC RECEVER
GPS
3.3v
J5
1
2
3
4
GPS_TX
GPS_RX
STM32F446RET6 Core[2C]
STM32F446RET6 Core[3B]
S4B-XH-A(LF)(SN)
GND
J4
1
2
3
+5V_SYS
SBUS_IN
EXTERNAL CONNECTORS[1C], I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE[3A], RC INPUT, STATUS
RC INPUT, STATUS LED, SWD HEADER[1A]
S3B-XH-A(LF)(SN)
GND
ESD OUTPUTS
J_M1
TIM2_CH1
STM32F446RET6 Core[2A]
S3B-XH-A(LF)(SN)
GND
J_M2
TIM2_CH2
STM32F446RET6 Core[2A]
S3B-XH-A(LF)(SN)
GND
J_M3
TIM2_CH3
STM32F446RET6 Core[2A]
S3B-XH-A(LF)(SN)
GND
J_M4
TIM2_CH4
STM32F446RET6 Core[2A]
S3B-XH-A(LF)(SN)
GND
LORA TELEMETRY
J7
LORA_M0
LORA_M1
LORA_TX
LORA_RX
LORA_AUX
STM32F446RET6 Core[2B]
STM32F446RET6 Core[2B]
STM32F446RET6 Core[2C]
STM32F446RET6 Core[2C]
STM32F446RET6 Core[2B]
+5V_SYS
EXTERNAL CONNECTORS[2A], I2C SENSORS B - BAROMETER, CURRENT MONITOR, BATTERY SENSE[3A], RC INPUT, STATUS LED, SWD HEADER[2A]
TSW-107-05-G-S
GND
C_LORA
100n
GND
Title
Size
A
Number
Revision
Date:
9/10/2026
Sheet of
File:
C:\Users\..\EXTERNAL CONNECTORS.SchDoc
Drawn By: